

# Onyx — ImageKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **ImageKit** (`SD:/lib/imagekit.so`, `user/Kits/imagekit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

## Contents

1. What it is
2. Using it
3. Index
4. `imagekit/imagekit.h`

## What it is

ImageKit is the system's one picture codec: BMP, GIF, PNG, JPEG, PCX and WebP read; PNG, JPEG, BMP and GIF written; resizing, turning, the photo adjustments.

|         |                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                             |
| Include | <code>#include "imagekit/imagekit.h"</code>                                                                                 |
| Link    | <code>lib/imagekit.imp.a</code>                                                                                             |
| Library | <code>SD:/lib/imagekit.so</code> — 46 entries in its table<br>( <code>user/Kits/imagekit/imagekit.abi</code> , append-only) |
| Sources | <code>user/Kits/imagekit/</code>                                                                                            |

## Using it

ImageKit is the system's one picture codec: BMP, GIF, PNG, JPEG, PCX and WebP read; PNG, JPEG, BMP and GIF written.

**A picture read, turned, saved in another format:**

```
#include "imagekit/imagekit.h"
```

```
ik_image *im = ik_load ("SD:/photos/cat.jpg", 0);
if (im)
{
    int w = ik_width (im), h = ik_height (im);
    unsigned *px = ik_pixels (im);           // w * h pixels, 0xAARRGGBB
    ik_rotate (im, 1);                       // a quarter turn clockwise
    ik_save (im, "SD:/photos/cat.png", 90);  // the format: the extension's
    ik_image_free (im);
}
```

**Pixels of the program's own, saved:**

```
ik_image *shot = ik_image_from (pixels, width, height, width);
ik_save (shot, "SD:/screenshots/shot.png", 0);
ik_image_free (shot);
```

**The formats, asked** — a file chooser's filter follows what the library knows:

```
struct ik_format f[16];
int n = ik_formats (f, 16);
```

Resizing (`ik_scale`), flipping (`ik_flip`), the photo adjustments and the EXIF orientation are in the same header.

## Index

Everything the headers declare, in their order — the details are in each header's part below.

| Name                          | What it does                                                 | Header                  |
|-------------------------------|--------------------------------------------------------------|-------------------------|
| <code>ik_free</code>          | a buffer the library returned                                | <code>imagekit.h</code> |
| <code>ik_image_new</code>     | transparent black; 0: no memory, a bad size                  | <code>imagekit.h</code> |
| <code>ik_image_from</code>    | a copy of pixels (stride: in pixels)                         | <code>imagekit.h</code> |
| <code>ik_image_copy</code>    | a new picture with the same pixels; 0: no picture, no memory | <code>imagekit.h</code> |
| <code>ik_image_free</code>    | the picture and its pixels freed (0: nothing)                | <code>imagekit.h</code> |
| <code>ik_width</code>         | in pixels (0: no picture)                                    | <code>imagekit.h</code> |
| <code>ik_height</code>        | in pixels (0: no picture)                                    | <code>imagekit.h</code> |
| <code>ik_format</code>        | IK_ARGB8                                                     | <code>imagekit.h</code> |
| <code>ik_pixels</code>        | its w * h pixels (until the image changes size)              | <code>imagekit.h</code> |
| <code>ik_fill</code>          | every pixel set to that 0xAARRGGBB value                     | <code>imagekit.h</code> |
| <code>ik_opaque</code>        | every pixel's alpha 255                                      | <code>imagekit.h</code> |
| <code>ik_flatten</code>       | laid on a colour, the result opaque                          | <code>imagekit.h</code> |
| <code>ik_has_alpha</code>     | 1: a pixel is not opaque                                     | <code>imagekit.h</code> |
| <code>ik_formats</code>       | how many there are (out: up to max)                          | <code>imagekit.h</code> |
| <code>ik_info</code>          | (a type)                                                     | <code>imagekit.h</code> |
| <code>ik_is_image_name</code> | by its extension (bmp gif png jpg jpeg jpe pcx webp)         | <code>imagekit.h</code> |
| <code>ik_probe</code>         | 1 / 0: not a picture this reads                              | <code>imagekit.h</code> |
| <code>ik_load</code>          | the picture (an animation: its first                         | <code>imagekit.h</code> |

| Name                            | What it does                                                                                | Header                  |
|---------------------------------|---------------------------------------------------------------------------------------------|-------------------------|
|                                 | frame); 0                                                                                   |                         |
| <code>ik_load_mem</code>        | the same from a file's n bytes in memory (flags: ignored, no orientation applied); 0        | <code>imagekit.h</code> |
| <code>ik_load_preview</code>    | the camera's preview, turned as seen; 0: none                                               | <code>imagekit.h</code> |
| <code>ik_load_format</code>     | the last picture read: "PNG", "JPEG"... ("": none)                                          | <code>imagekit.h</code> |
| <code>ik_frames_load</code>     | the file read, every frame decoded (a still picture: one frame); 0: not read                | <code>imagekit.h</code> |
| <code>ik_frames_load_mem</code> | the same from a file's n bytes in memory; 0                                                 | <code>imagekit.h</code> |
| <code>ik_frames_count</code>    | how many frames it holds (0: none)                                                          | <code>imagekit.h</code> |
| <code>ik_frames_pixels</code>   | frame i: w * h pixels (the library's)                                                       | <code>imagekit.h</code> |
| <code>ik_frames_delay</code>    | how long it shows, ms (0: a still picture)                                                  | <code>imagekit.h</code> |
| <code>ik_frames_width</code>    | every frame's width, pixels                                                                 | <code>imagekit.h</code> |
| <code>ik_frames_height</code>   | every frame's height, pixels                                                                | <code>imagekit.h</code> |
| <code>ik_frames_free</code>     | the frames it still holds and the handle freed (0: nothing)                                 | <code>imagekit.h</code> |
| <code>ik_frames_take</code>     | The frames handed over to the caller (px[i]                                                 | <code>imagekit.h</code> |
| <code>ik_inflate</code>         | A deflate stream inflated (zlib                                                             | <code>imagekit.h</code> |
| <code>ik_encode</code>          | 0 / -1                                                                                      | <code>imagekit.h</code> |
| <code>ik_save</code>            | the format from the path's extension → 0 / -1                                               | <code>imagekit.h</code> |
| <code>ik_encode_pixels</code>   | Raw pixels encoded (stride                                                                  | <code>imagekit.h</code> |
| <code>ik_scale</code>           | The low-level one                                                                           | <code>imagekit.h</code> |
| <code>ik_scale_rgb</code>       | The same for pixels whose top byte is not an alpha (0x00RRGGBB, a window's canvas, a photo) | <code>imagekit.h</code> |

| Name                         | What it does                                                      | Header                  |
|------------------------------|-------------------------------------------------------------------|-------------------------|
| <code>ik_resize</code>       | a new picture of that size                                        | <code>imagekit.h</code> |
| <code>ik_fit</code>          | within the box, proportions kept (grow 0: never larger)           | <code>imagekit.h</code> |
| <code>ik_cover</code>        | the middle cut to that shape and brought to it (a thumbnail)      | <code>imagekit.h</code> |
| <code>ik_crop</code>         | in place → 0 / -1                                                 | <code>imagekit.h</code> |
| <code>ik_rotate</code>       | clockwise, in place                                               | <code>imagekit.h</code> |
| <code>ik_flip</code>         | 0: left-right, 1: top-bottom                                      | <code>imagekit.h</code> |
| <code>ik_orient</code>       | 1..8 applied                                                      | <code>imagekit.h</code> |
| <code>ik_straighten</code>   | a small angle, enlarged so no corner is empty (the result opaque) | <code>imagekit.h</code> |
| <code>ik_adjust</code>       | (a type)                                                          | <code>imagekit.h</code> |
| <code>ik_adjust_apply</code> | in place (the alpha kept)                                         | <code>imagekit.h</code> |
| <code>ik_adjust_auto</code>  | "enhance": from the picture's histogram                           | <code>imagekit.h</code> |
| <code>ik_filter_name</code>  | "Black and white" ...                                             | <code>imagekit.h</code> |

## imagekit/imagekit.h

imagekit.h -- ImageKit: pictures, one copy for the whole system (SD:/lib/imagekit.so; the shared libraries: docs/03 section 5.6). A program links lib/imagekit.imp.a and calls plain C functions (ik\_\*). The first version (2026-10-05) is the base every program that shows or writes a picture needs; the layers, the masks and the brushes of the photo editor come on top of it later.

```

images      a picture in memory (ik_image): w x h pixels 0xAARRGGBB, straight
alpha, rows one
            after the other
  reading    BMP, GIF (animated too), PNG, JPEG, PCX, WebP -- from a file or from
memory; what a
            file says without decoding it (its size, the camera's orientation,
date, model);
            the camera's own preview
  writing     PNG (zlib's compression, through FileKit), JPEG, BMP, GIF -- to a file
or to memory
  transforms ONE resize, right in alpha (the average of what a pixel covers when
smaller,
            bilinear when larger); fit, cover, crop, quarter turns, flips, the
camera's
            orientation applied, straightened by a small angle
  adjustments exposure, contrast, highlights, shadows, saturation, warmth, sharpness
and six
            filters as one tone curve; "enhance" from the histogram (Photos' own)

```

Every call takes integers and pointers only: a program built without the FPU calls it. An `ik_image` is opaque (its format may grow: 16 bits a channel later); a buffer the library returns (`void **out`) is freed with `ik_free`.

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```
void ik_free (void *p);           // a buffer the library returned
```

## a picture in memory

```
#define IK_ARGB8    1           // 0xAARRGGBB, 8 bits a channel, straight alpha (the only format today)
```

```
typedef struct ik_image ik_image;
ik_image *ik_image_new (int w, int h);           // transparent black; 0: no memory, a bad size
ik_image *ik_image_from (const unsigned *px, int w, int h, int stride); // a copy of pixels (stride: in pixels)
ik_image *ik_image_copy (const ik_image *im);    // a new picture with the same pixels; 0: no picture, no memory
void ik_image_free (ik_image *im);               // the picture and its pixels freed (0: nothing)
int ik_width (const ik_image *im);              // in pixels (0: no picture)
int ik_height (const ik_image *im);             // in pixels (0: no picture)
int ik_format (const ik_image *im);             // IK_ARGB8
unsigned *ik_pixels (ik_image *im);             // its w * h pixels (until the image changes size)
void ik_fill (ik_image *im, unsigned argb);     // every pixel set to that 0xAARRGGBB value
void ik_opaque (ik_image *im);                  // every pixel's alpha 255
void ik_flatten (ik_image *im, unsigned rgb);   // laid on a colour, the result opaque
int ik_has_alpha (const ik_image *im);         // 1: a pixel is not opaque
```

## the formats, asked (version 2)

A program lists what the library reads and writes instead of keeping its own list: a format the library learns is one every program has.

```
struct ik_format
{
    char name[12];           // "PNG", "JPEG", "GIF", "BMP", "PCX", "WebP"
    char extensions[40];     // "jpg jpeg jpe" (lower case, a space between)
    int can_read, can_write;
    int alpha;              // it keeps transparency
    int animated;           // it may hold several frames (read as ik_frames)
    int reserved[4];
};
int ik_formats (struct ik_format *out, int max); // how many there are (out: up to max)
```

## reading

```
#define IK_ORIENT 1 // the camera's orientation (EXIF) applied: as the
photo is seen

struct ik_info
{
    int w, h; // as stored (before the orientation); 0: not known
    without decoding
    int orientation; // 1..8 (EXIF), 1: as stored
    long long taken; // when (seconds, local time, from 1970), 0: not
known
    char camera[64]; // "Google Pixel 8" ("": not known)
    char exposure[64]; // "f/1.7 · 1/640 s · ISO 50 · 6.9 mm"
    int has_preview; // the camera's own small picture is in the file
    (ik_load_preview)
    int reserved[8];
};
int ik_is_image_name (const char *name); // by its extension (bmp gif png
jpg jpeg jpe pcx webp)
int ik_probe (const char *path, struct ik_info *out); // 1 / 0: not a picture
this reads
ik_image *ik_load (const char *path, int flags); // the picture (an animation:
its first frame); 0
ik_image *ik_load_mem (const void *data, unsigned n, int flags); // the same from
a file's n bytes in memory (flags: ignored, no orientation applied); 0
ik_image *ik_load_preview (const char *path); // the camera's preview,
turned as seen; 0: none
const char *ik_load_format (void); // the last picture read: "PNG",
"JPEG"... ("": none)
```

An animation (a GIF): every frame.

```
typedef struct ik_frames ik_frames;
ik_frames *ik_frames_load (const char *path); // the file read, every frame decoded
(a still picture: one frame); 0: not read
ik_frames *ik_frames_load_mem (const void *data, unsigned n); // the same from a
file's n bytes in memory; 0
int ik_frames_count (const ik_frames *f); // how many frames it holds (0: none)
const unsigned *ik_frames_pixels (const ik_frames *f, int i); // frame i: w * h
pixels (the library's)
int ik_frames_delay (const ik_frames *f, int i); // how long it shows, ms (0:
a still picture)
int ik_frames_width (const ik_frames *f); // every frame's width, pixels
int ik_frames_height (const ik_frames *f); // every frame's height, pixels
void ik_frames_free (ik_frames *f); // the frames it still holds and the handle freed
(0: nothing)
```

The frames handed over to the caller (px[i]: new unsigned[w \* h], the caller's to delete []; delay[i]: ms): up to max of them → how many; f keeps none of those. (UIKit's img\_load is this.)

```
int ik_frames_take (ik_frames *f, unsigned **px, int *delay, int max);
```

A deflate stream inflated (zlib: with its header, as PNG's; else raw): a new unsigned char[] of \*out\_n bytes (the caller's to delete []), or 0. (UIKit's img\_inflate; a new program takes FileKit's.)

```
unsigned char *ik_inflate (const void *data, unsigned n, int zlib, unsigned *out_n);
```

## writing

format: "png", "jpg" / "jpeg", "bmp", "gif" (letters' case ignored). quality: JPEG's 1..100 (0: 90). A picture with transparency keeps it in PNG; GIF keeps its clear pixels (alpha < 128) clear and the others opaque; JPEG and BMP lay it on white.

```
int ik_encode (const ik_image *im, const char *format, int quality, void **out,
unsigned *out_n);    // 0 / -1
int ik_save (const ik_image *im, const char *path, int quality);    // the format
from the path's extension -> 0 / -1
```

Raw pixels encoded (stride: pixels a row). alpha: 1 kept where the format has it (PNG: RGBA), 0 not (PNG: RGB, whatever the top bytes hold), -1 kept when a pixel is not opaque. (img/pngsave.hpp's encoders are this, in the programs built with PNGSAVE\_USE\_IMAGEKIT.)

```
int ik_encode_pixels (const unsigned *px, int w, int h, int stride, const char
*format, int quality, int alpha,
void **out, unsigned *out_n);
```

## transforms

The low-level one: the rectangle (sx, sy, sw, sh) of src (src\_w x src\_h, src\_stride pixels a row) brought to dst (dw x dh, dst\_stride pixels a row). Right in alpha (the colours weighed by it). A big shrink reads at most  $\sim 8 \times 8$  of the pixels each result covers (a 24 Mpx photo's thumbnail stays quick).

```
void ik_scale (const unsigned *src, int src_w, int src_h, int src_stride, int sx, int
sy, int sw, int sh,
unsigned *dst, int dst_stride, int dw, int dh);
```

The same for pixels whose top byte is not an alpha (0x00RRGGBB, a window's canvas, a photo): the colours alone; the result's top byte is 0.

```
void ik_scale_rgb (const unsigned *src, int src_w, int src_h, int src_stride, int sx,
int sy, int sw, int sh,
unsigned *dst, int dst_stride, int dw, int dh);
ik_image *ik_resize (const ik_image *im, int w, int h);    // a new picture of that
size
ik_image *ik_fit (const ik_image *im, int max_w, int max_h, int grow);    // within the
box, proportions kept (grow 0: never larger)
ik_image *ik_cover (const ik_image *im, int w, int h);    // the middle cut to that
shape and brought to it (a thumbnail)
int ik_crop (ik_image *im, int x, int y, int w, int h);    // in place -> 0 / -1
int ik_rotate (ik_image *im, int quarter_turns);    // clockwise, in place
int ik_flip (ik_image *im, int vertical);    // 0: left-right, 1: top-bottom
int ik_orient (ik_image *im, int exif_orientation);    // 1..8 applied
int ik_straighten (ik_image *im, int millidegrees);    // a small angle, enlarged so
no corner is empty (the result opaque)
```

## adjustments (Photos')

```
#define IK_FILTER_NONE    0
#define IK_FILTER_BW    1
#define IK_FILTER_WARM    2
#define IK_FILTER_COOL    3
#define IK_FILTER_VINTAGE    4
#define IK_FILTER_VIVID    5
#define IK_FILTERS    6
```

```
struct ik_adjust
```

```

{
    int exposure, contrast, highlights, shadows, saturation, warmth;    // -100..100,
0: as it is
    int sharpness;                                                    // 0..100
    int filter;                                                        // IK_FILTER_*
    int reserved[8];                                                  // 0
};
void ik_adjust_apply (ik_image *im, const struct ik_adjust *a); // in place (the
alpha kept)
void ik_adjust_auto (const ik_image *im, struct ik_adjust *out);    // "enhance":
from the picture's histogram
const char *ik_filter_name (int filter);                            // "Black and white" ...
}
#endif

```